

connectivity protocols before moving towards the edge for secure testing.

Sensor ecosystem for wireless connectivity

First, a rugged testing bench for the hardware needs to be created that directly interfaces with the diagnosis system of a car for capturing data (people can use DSP and microcontrollers). Second, for software compatibility of the hardware, firmware-over-the-air (FOTA) needs to be implemented to update data within the hardware itself (while carrying out data and embedded communications).

Coming to wireless connectivity, the best way for validating a connected car is through edge computing. It can be entirely within the car (as a hardware part) or on the cloud for data transfer and acquisition. On the sensing side, smart sensors need to be implemented. For instance, face recognition sensors can capture and process raw data within the camera, making them intelligent as they are not dependent on any external hardware for encoding and decoding.

“If we put AI on the edge and then perform bi-directional communication (wired or wireless), then the sensor can also benefit from it, making it a smart sensor,” says Dharmendra Kumar, IoT head, Arihant Electricals.

When vehicle systems relay sensor-generated data, drivers can receive alerts about road conditions and driving hazards, such as congested roads and highways, well before encountering these problems. When vehicle systems are connected to the roadway infrastructure, sensor-generated data can supply accurate, real-time traffic data, allowing mapping programs to plot the most efficient route, so that drivers save time and greenhouse gas emissions are minimised.

One of the biggest challenges is on the mechanical side. For vibration resistance, especially in the wheels, shock-proof sensors need to be developed. Another challenge is the lack of ecosystems due to which car companies do most of the validation for connected cars. Although a car manufacturer can create a good car, it is difficult to go for IoT and edge computing at the same time. Therefore, IoT engineers should be invited to validate them. This is a good way to create an ecosystem.

The discussion further opens up critical thoughts on the impact of add-on sensors and actuators benefitting the verification and validation process of systems in vehicles.

Deep learning to avoid chaos

Machine learning (or deep learning) is a foolproof technique for end-to-end driving in the real world and ensuring a vehicle operates well while identifying the issues. But what if something is not present in the training data? Will this technique be able to solve abrupt real-world problems?

There are two possible standard solutions:

1. Use simulations. Accurate 3D mapping of everything is important. Testing hundreds and thousands of kilometre cycles in the physical world takes over a year. However, this can be easily achieved with a simulation program overnight. But then simulators assume people drive in a civilised way, which is a bad assumption. What do you do

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